IMO – the International Maritime Organization

What it is, What it does, How it works - particular focus on FSI and PSC

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IMO – International Maritime Organization

What it is, What it does, How it works





IMO mission:
safe, secure and efficient shipping on clean oceans



INTERNATIONAL MARITIME ORGANIZATION

IMO

- Specialised UN agency
- Headquarters in UK since 1958
- Annual budget £30+ million
- Secretariat – 300 staff, more than 50 nationalities



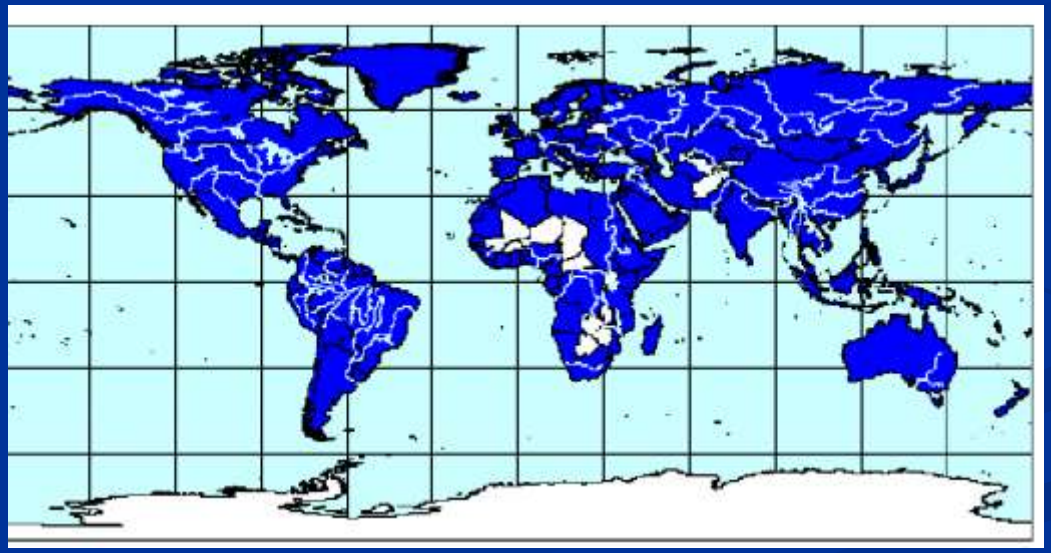
Ten largestcontributors to IMO in 2013 Assessed contributions based on flat base rate with additional components based on ability to pay and merchant fleet tonnage

	Amount GBP	% of total
1 Panama	5,519,785	18.29
2 Liberia	3,212,409	10.65
3 Marshall Islands	2,031,796	6.74
4 Singapore	1,460,995	4.84
5 United Kingdom	1,426,952	4.73
6 Bahamas	1,345,313	4.46
7 China	1,103,219	3.66
8 Malta	1,091,162	3.62
9 Greece	1,079,047	3.58
10 Japan	1,010,345	3.35

IMO - global coverage

170 Member States, three associate members

IGOs and NGOs participate as observers



IMO at work

- Assembly
- Council – 40 Members

- Committees:
- Maritime Safety Committee (MSC)
- Marine Environment Protection Committee (MEPC)
- Legal Committee (LEG)
- Facilitation Committee (FAL)
- Technical Cooperation Committee (TCC)

Sub-Committees

- Sub-Committee on Ship Design and Construction (SDC)
- Sub-Committee on Pollution Prevention and and Response (PPR)
- Sub-Committee on Human Element, Training and Watchkeeping (HTW)
- Sub-Committee on Ship Systems and Equipment (SSE)
- Sub-Committee on Navigation, Communication and Search
- Cargoes and Containers (CCC)
- Sub-Committee on Implementation of IMO Instruments (III)





Progress of measures at IMO



- **Proposals for new, or amendments to existing, mandatory instruments**

- **-** a **compelling need** for such amendments should be demonstrated by the proponent(s), and an analysis of the implications of such amendments, particularly those with far-reaching implications and consequential proposals for other amendments, having regard to the costs to the maritime industry, the legislative and administrative burdens involved and benefits which would accrue therefrom, should be provided……

Application to real ships

	Parties	World Tonnage
SOLAS	162	99.20%
Load lines	161	99.19%
MARPOL	152	99.20%
COLREGS	155	98.71%
STCW	156	99.22%

IMO instruments



- Some 50 IMO Conventions and Protocols
- Hundreds of codes, guidelines and recommendations
- Almost every aspect of shipping covered:
  - Design
  - Construction
  - Equipment
  - Maintenance
  - Navigation
  - Communication
  - SAR
  - Cargoes
  - Crew
  - Security
  - Pollution prevention

Implementation – whose role?

- Flag States on own ship
  - classification societies
  - voluntary audit scheme

    first audits 2006; now moving to mandatory scheme



- Port State Control
- IMO – no “policing” mandate

EFFECTS OF FLAG STATE PERFORMANCE ON SHIPPING COMPANIES

- Shipping companies have primary responsibility for the safe operation of their ships and welfare of crew.
- However Government has a crucial role to play with regard to:
  - Implementing important conventions
  - Enforcing their provisions nationally

States have rights and obligations to control ships

- those registered under their flag ; and
- those arriving in their ports
- in order to verify and ensure that ships are in compliance with
  - applicable standards, national and international for flag States and
  - international for port States

FS surveys and PS inspections

- FS Surveys:
  - Regular intervals
  - Defined scope
  - Measures – rectify deficiency in a defined period, FS comments on detention,..
- PS inspections
  - Random selection
  - Not full survey – selection of areas for inspection (initial, more detailed, expanded)
  - Measures - rectify deficiency in a defined period, PS detention, banning,..

The Interactions between FS and PS

- PSC as “second line of defence”

- To fulfil its international obligations it is essential for a flag State to establish and maintain an effective control over the ships flying its flag. The need for this control is set out in Article 94 of UNCLOS and implied in the IMO Conventions

“genuine link”

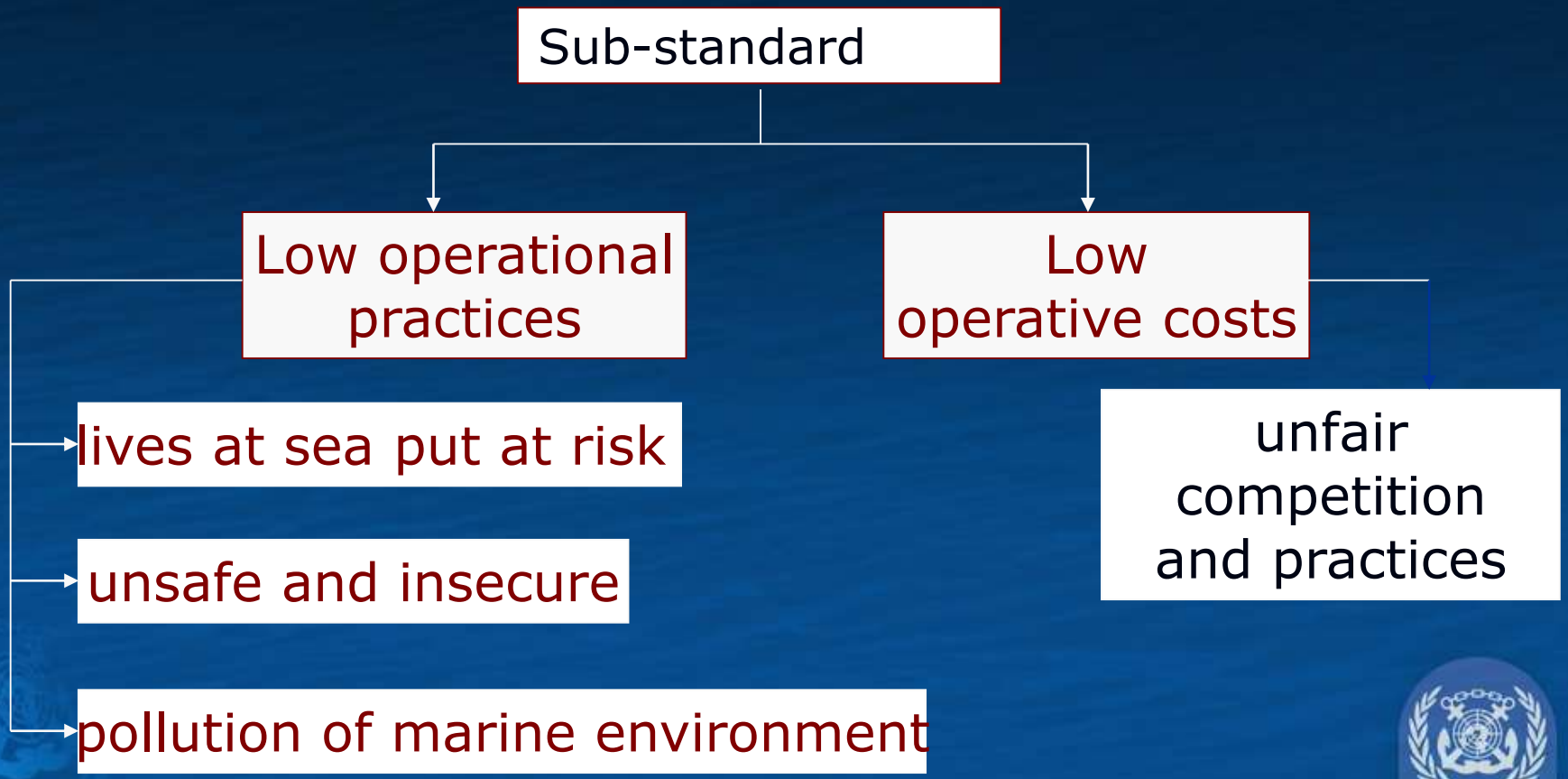
The Interactions between FS and PS

- Exchange of data through IMO:
  - Flag States contact points for PSC
  - Flag States comments on PS detentions
  - Harmonization of codes
  - GISIS (PSC module to come)
- Exchange of data directly:
  - PS:
    - Detention report (data basis for all other inspections)
    - Information to and Consultation with FS
  - FS:
    - Confirmation to follow-up on PS inspection results

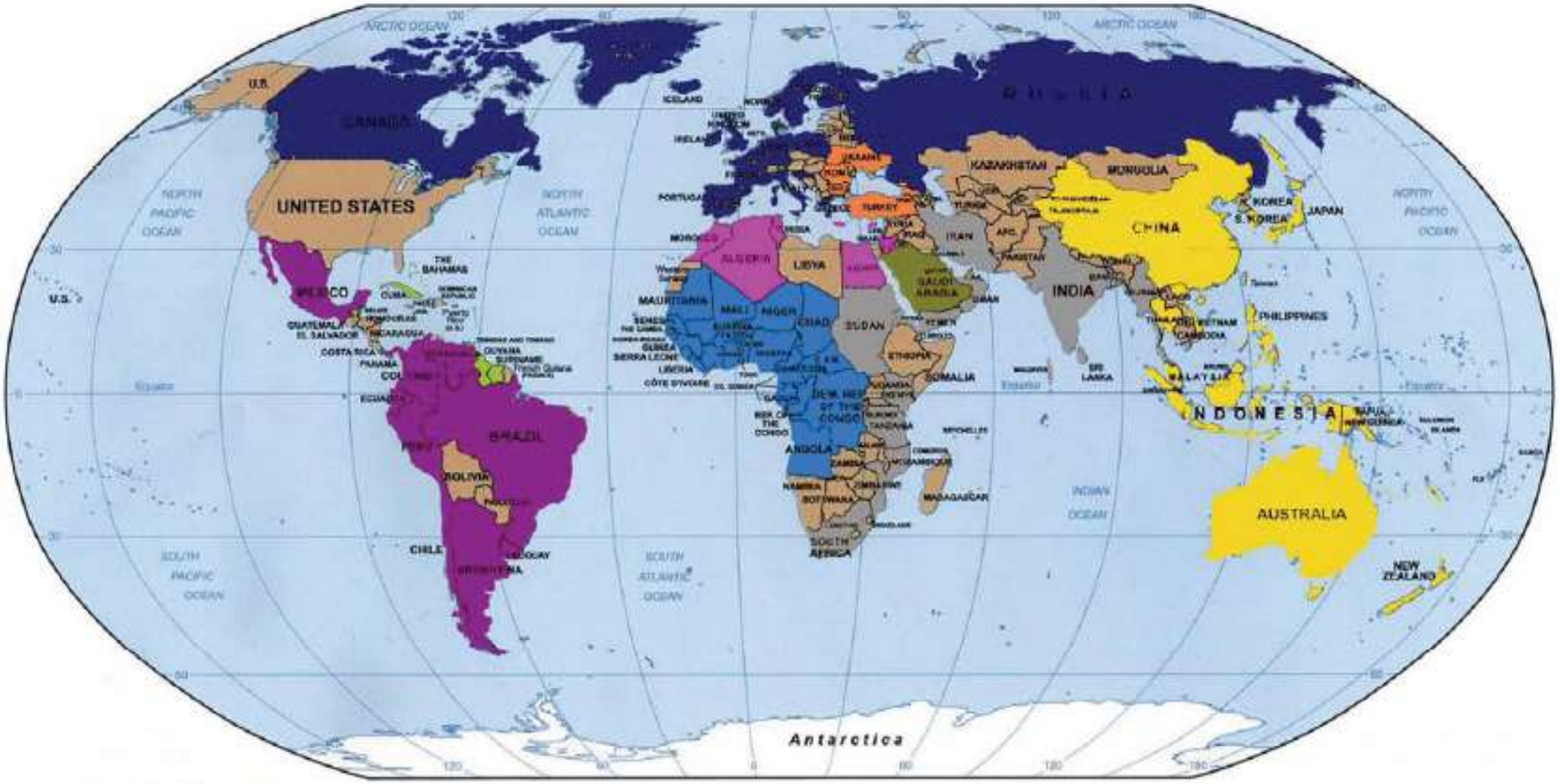
- FS as a Party to conventions has the primary responsibility to monitor its fleet
  - Register ships
  - Implement conventions in national laws
  - Perform statutory certification of ships
    - Carry out periodical surveys
    - Issue certificates

Port State control (PSC)

to be considered from the point of view of appropriate procedures, activities and practices with the goal of eradicating the operation of sub-standard ships.



World-Wide Port State Control



Abuja	Black Sea	Caribbean	Indian	Med	Paris	Riyadh	Tokyo	Vina Del Mar

Safety Regulations for non-Convention ships (GlobalRegs)

- ship safety certification and safe manning.
- approval of marine equipment.
- registration of persons on board passenger ships.
- Safety for cargo ships of a more than 500 gross tonnage and for passenger ships of more than 24 metres in length.
- Safety for passenger ships of less than 24 metres in length.
- Safety for non-SOLAS cargo ships of more than 12 metres in length.
- Safety for ships engaged in inland waterways.
- Safety for fishing vessels of less than 12 metres; of 12 metres in length and above, but less than 24 metres; and of more than 24 metres in length.
-Surveyors' Note Book.

available, in English, on the IMODOCS webpage
Set of standards regulations on non-Convention ships (GlobalReg))

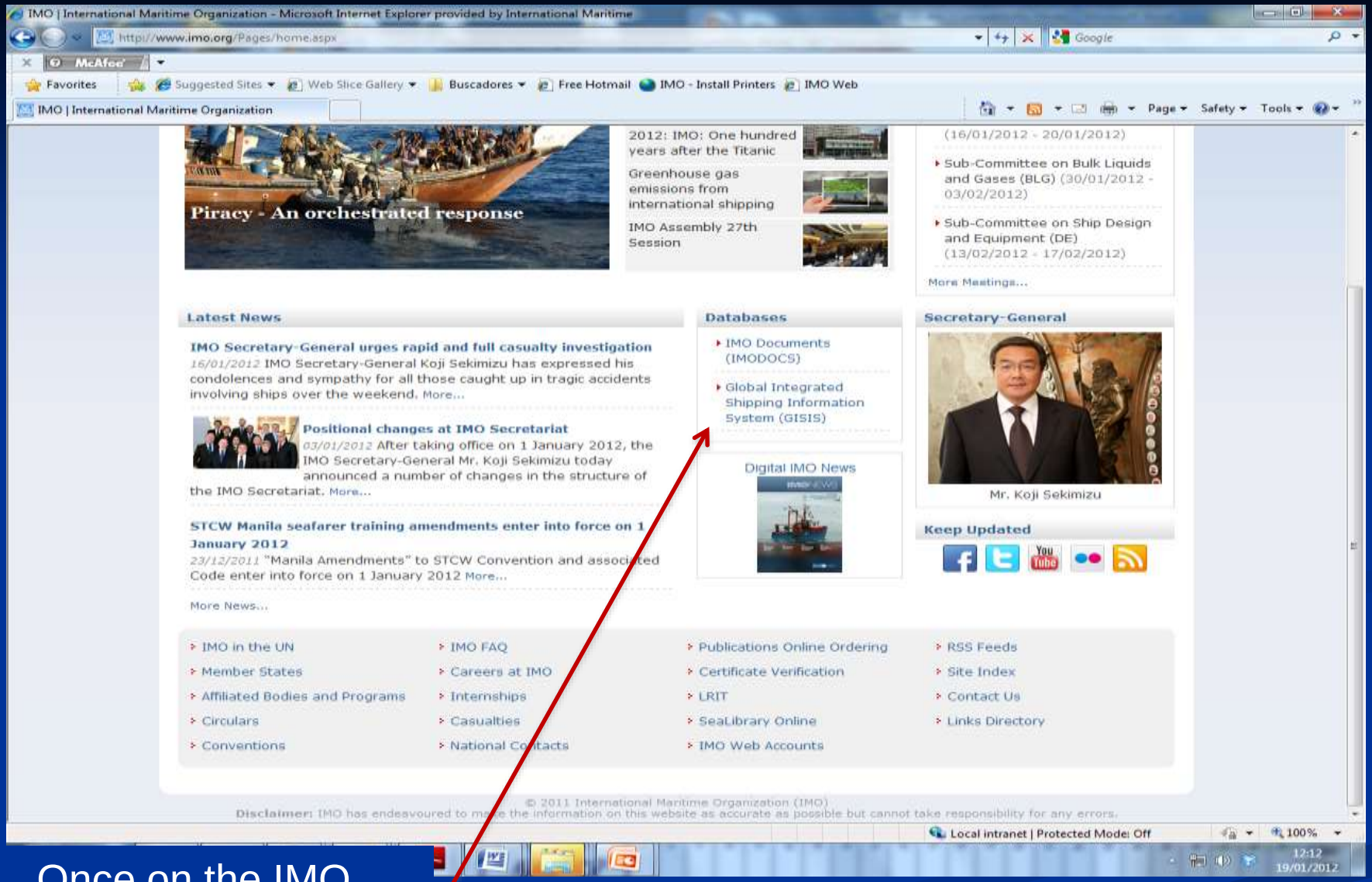
Methodology for the processing of PSC data and its use

- EQUASIS
- Global Integrated Shipping Information System (GISIS)
- launched in mid-2005

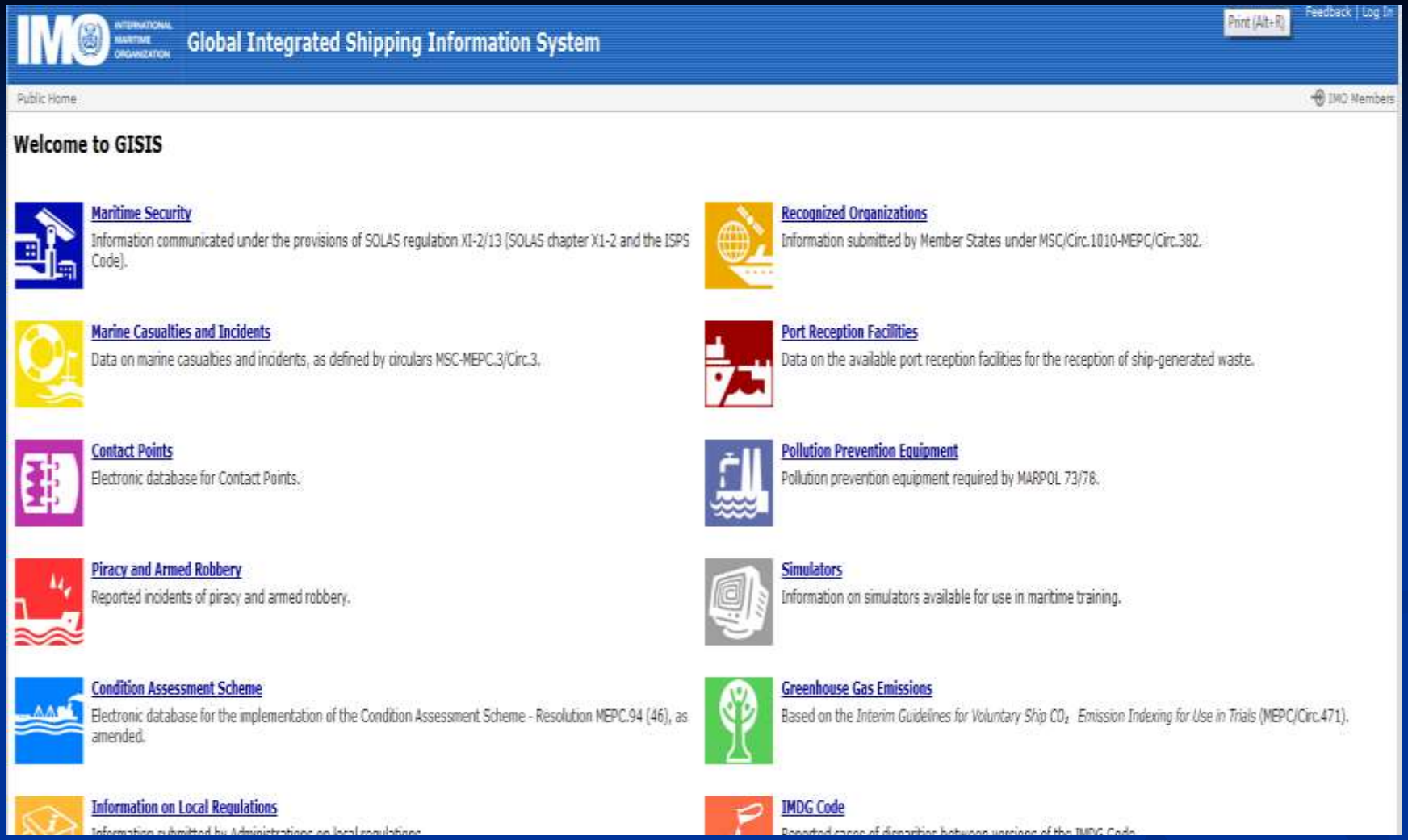
To compile and make available, on a world-wide basis, appropriate and accurate information on individual ships and groups of ships relevant to maritime safety, security and pollution prevention and the safeguarding of adequate working and living conditions of seafarers.
(under reporting requirements contained in international instruments)

GISIS

- The Global Integrated Shipping Information System (GISIS) presently consists of 22 modules, with a further six under development, for the collection, processing and sharing of shipping-related data in order to assist Member States and the Secretariat in carrying out their respective and complementary duties, generate reports and provide information about shipping to the public. GISIS has over 70,000 public-registered users with an estimated 4,900 pages viewed per day and 9,000 visits per month. Users, with authorized access to the GISIS Members' Area, make approximately 1,300 visits per month.



Once on the IMO Website, Click on GISIS


Global Integrated Shipping Information System
Print (Alt+R)
Feedback | Log In
Public Home
IMO Members
Welcome to GISIS
Maritime Security
Information communicated under the provisions of SOLAS regulation XI-2/13 (SOLAS chapter X1-2 and the ISPS Code).
Recognized Organizations
Information submitted by Member States under MSC/Circ.1010-MEPC/Circ.382.
Marine Casualties and Incidents
Data on marine casualties and incidents, as defined by circulars MSC-MEPC.3/Circ.3.
Port Reception Facilities
Data on the available port reception facilities for the reception of ship-generated waste.
Contact Points
Electronic database for Contact Points.
Pollution Prevention Equipment
Pollution prevention equipment required by MARPOL 73/78.
Piracy and Armed Robbery
Reported incidents of piracy and armed robbery.
Simulators
Information on simulators available for use in maritime training.
Condition Assessment Scheme
Electronic database for the implementation of the Condition Assessment Scheme - Resolution MEPC.94 (46), as amended.
Greenhouse Gas Emissions
Based on the Interim Guidelines for Voluntary Ship CO2 Emission Indexing for Use in Trials (MEPC/Circ.471).
Information on Local Regulations
IMDG Code

None performance of flag States

- Could lead to:
  - Port state control targeting national fleet leading to greater number of inspections
  - Unnecessary delays in ports
  - greater potential for penalties
  - PSC inspectors being more inclined to make issue of non-critical deficiencies

SELF ASSESSMENT OF FLAG STATE PERMANCE

- Assembly Resolution A881 (21)
- To assess level of implementation of IMO instruments and identify areas of difficulty particularly in relation to technical assistance

PROBLEMS WITH RESOLUTION A 881(21)

- Submission of assessment at discretion of members
- Reports are confidential and can only be released with consent of submitting Govt
- No verification

IMO MODEL AUDIT SCHEME

- The Council decided to develop a model audit scheme and a Code for its implementation
- 90th session of Council in June 2003 approved proposed objectives and principles.

IMO MODEL AUDIT SCHEME

- 23rd session of Assembly, November 2003
- Adopted resolution A946 (23) - Voluntary IMO Member State Audit, endorsing the decision to develop a model audit scheme IN SUCH A MANNER AS NOT TO EXCLUDE THE POSSIBILITY IN FUTURE OF IT BECOMING MANDATORY

IMO AUDIT SCHEME

- 28th session of Assembly, November 2013
- Adopted resolution A1067 (28) – Framework and Procedures for the IMO Member State Audit Scheme
- Adopted resolution A1068 (28) – Transition from the Voluntary IMO Member State Audit Scheme to the IMO Member State Audit Scheme

CODE FOR IMPLEMENTATION OF IMO INSTRUMENTS (Res. A.1070(28))

- The code was developed to form the basis of the audit standard and has identified all relevant obligations of Parties to IMO instruments
- SCOPE:
  - 1974 SOLAS and 1978 Protocol, as amended
  - MARPOL 73/78, as amended
  - STCW 1978, as amended
  - LOADLINES 1966, as modified by the 1988 Protocol
  - TONNAGE 1969
  - COLREG 1972

CODE FOR IMPLEMENTATION OF IMO INSTRUMENTS (Res. A.1070(28))

III Code (Res. A.1070(28))

Resolution A.1077-Obligations under III Code

- Obligations of Contracting Governments/Parties
- Specific Flag State Obligations

- Specific Coastal State Obligations
- Specific Port State Obligations

Delegation of work

- need for delegation

- virtually all Administrations find it necessary to delegate some or all of the survey and certification work associated with the international conventions on safety and the prevention of pollution.
- the Conventions require authority to be delegated only to "recognized organizations" which, in practice, are certain classification societies.
- delegation arises from the Administration's need for:

  .1 the in-depth technical expertise possessed by the established classification societies;

  .2 the world-wide service which the societies can provide; and

  .3 avoiding duplication of surveys and inspections and the consequent costs to the shipowner.

VERIFICATION AND MONITORING

- CLEAR PROCEDURE FOR COMMUNICATION AND REPORTING
- REGULAR REPORTING TO ADMINISTRATION
- ADDITIONAL SHIP INSPECTION BY ADMINISTRATION
- EVALUATION OF CERTIFICATES BY ADMINISTRATION
- PERIODIC REVIEW OF THE ORGANIZATION'S QUALITY SYSTEM

CODE FOR RECOGNIZED ORGANISATIONS (RO CODE)

- **RESOLUTION MSC.349(92) – adopted 06/2013**
- **The code was developed on the basis of *A.739(18)* - Guidelines for the Authorization of Organizations acting on behalf of the Administration; and**
- **2006 amendments to the Guidelines**

What is IMO ?

Safer shipping

Cleaner oceans

"IMO: SAFE, SECURE AND EFFICIENT SHIPPING ON CLEAN OCEANS"

IMO Technical co-operation



- Resource imbalance
- Needs assessment
- Donors – expertise, funding, training
- World Maritime University, IMLI





IMO World Maritime Day theme 2014



IMO

WORLD MARITIME DAY 2014

CONVENTIONS

EFFECTIVE IMPLEMENTATION

Maritime security



- **Maritime security measures**

(adopted 2002 – in force July 2004)

- **ISPS Code**
  - **Risk assessment – level of threat**
  - **Ship and port facility security officers**
  - **Security assessments, plans and certificates**
  - **RSOs**

- **Other SOLAS amendments**
  - **Ship security alerts, accelerated AIS introduction,**
  - **LRIT**

1988 SUA Convention

- Seizing/control of ships
- Acts of violence against person on board
- Destruction or damage to ship or cargo
- Placement of devices or substances to destroy or damage ship
- Damage/destruction of maritime navigational facilities
- False information

Revision of SUA Convention



Revision of SUA Convention

- New Protocols adopted 10/2005 – in force 07/2010
- Inclusion of new offences - carriage of WMD (i.e. nuclear & fissile - dual use materials)
- Linkage with non-proliferation treaties
- Inclusion of provisions for boarding of suspect ships on the high seas



Revision of SUA Convention:
Article 8 *bis* Boarding provisions

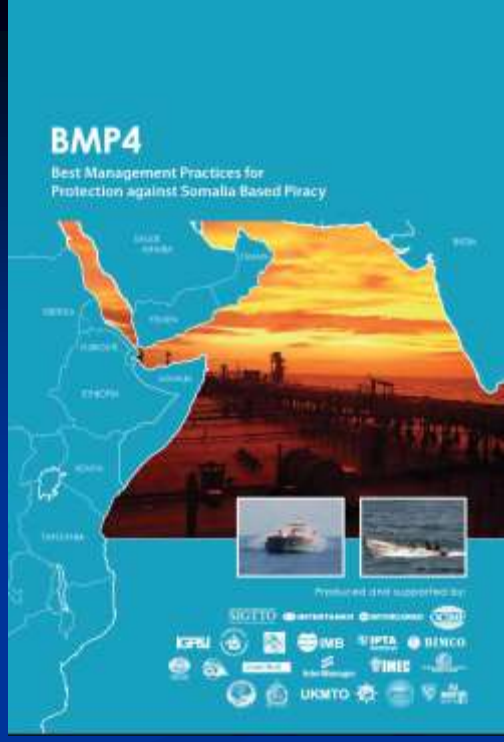

Launch of the Action Plan to promote
the 2011 World Maritime Day theme:
PIRACY
IMO
ORCHESTRATING
THE RESPONSE
IMO Headquarters
Thursday, 3 February 2011 at 11.15 a.m.
IMO
INTERNATIONAL
MARITIME
ORGANIZATION

Piracy

- Piracy off West Africa in 1985
- Piracy in Southeast Asia and Malacca Strait early 90s
- Piracy off Somalia as from 2005
- Originally - protecting persons at sea, WFP ships and vital shipping lanes
- Various IMO and UN Security Council resolutions
- Piracy off west and central

  Africa is a growing problem

IMO Guidance



- Global guidance to Governments, ship owners, ship operators and crews on suppression of piracy
- Investigation of offences
- Somalia-specific guidance including BMP
- Guidance to flag States, coastal and port States, ship-owners, ship operators and ship masters on privately contracted armed security personnel (PCASP)
- Guidance to PCASPs
- Guidance for Private Maritime Security Companies (PMSC) providing PCASPs

Regional Agreements

- South East Asia and Malacca Straits
  - ReCAAP 2006
  - Cooperative Mechanism 2007
- West and Central Africa
  - West and Central Africa Code of Conduct (WCACoC)
- East Africa (Somalia)
  - Djibouti Code of Conduct (DCoC)
  - Kampala process and successor

ENSURING SAFE, SECURE AND RELIABLE SHIPPING IN THE ARCTIC OCEAN

DEVELOPMENT OF THE POLAR CODE







IMO INTERNATIONAL MARITIME ORGANIZATION

IMO and Goal-Based Standards,

including implications for passenger ship safety





INTERNATIONAL MARITIME ORGANIZATION

Goal-Based Approach



IMO path to Goal Based regulations

Prescriptive Approach

- Presecriptive regulations tend to represent past experience.
- Become less and less relevent over time.
- Can hold back innovative ship designers
- Makes it difficult to move quickly to meet challenges
- Regulatory process is generally slow in adopting new regulations

New Technology

- Today's technology, makes easier to meet future challengers.
- In the 1990s, IMO Members start expressing concerns that presciptive regulatory approach is holding back innovation.
- Recognized by IMO Maritime Safety Committee (MSC).
- MSC starts to incorporate goal-based regulations in SOLAS & related codes.

Goal-Based Approach

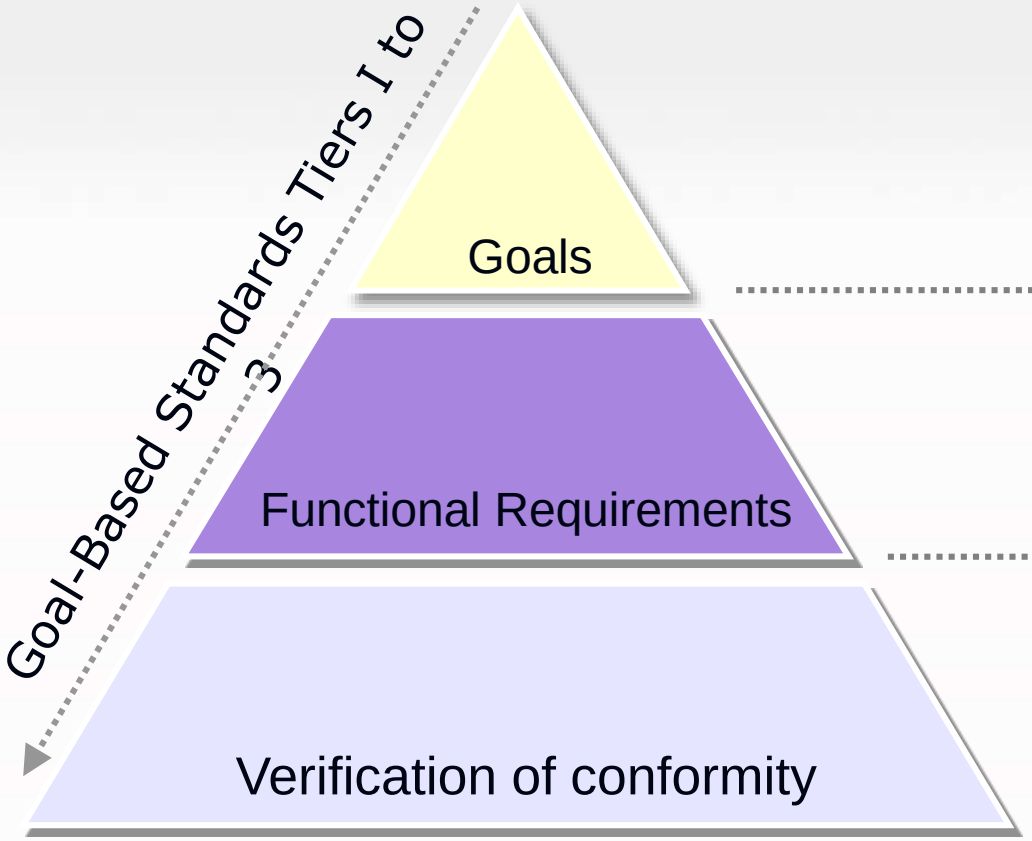
- One of the first goal-based initiatives was undertaken with revision of SOLAS chapter II-2.
- New SOLAS regulation II-2/17 on alternative design & arrangements.
- Passenger ship safety initiative prepares goal-oriented regulations.
- Althernative design regulations now also in SOLAS chapters II-1 and III

IMO INTERNATIONAL MARITIME ORGANIZATION

GBS – Tiers developed by IMO



The first three tiers are the focus of IMO and this presentation



- Tier I: The High-level objectives.
- Tier II: Criteria to be satisfied in order to conform to the goals.
- Tier III: Procedures for verifying that rules conform to goals and functional requirements.

IMO and FORMAL SAFETY ASSESSMENT (FSA)





INTERNATIONAL MARITIME ORGANIZATION

Passenger ship safety

- Titanic – first SOLAS adopted in 2014
- 1929, 1948, 1960
- 1974 version – introduced “tacit acceptance”

- Accidents have led to revisions, also revisions due to “what might happen”, 2006 passenger ship amendments



- Time for a new SOLAS? SOLAS 2024?

Fishing vessel safety

- Torremolinos Protocol 1993 - South Africa conference October 2012 adopted Cape Town 2012 agreement on implementation
- STCW-F 1995 enters into force 29 September 2012
- Thousands of lives lost annually in fishing sector worldwide
- IMO/ILO/FAO: joint work on IUU fishing and guidance e.g. Code of Safety for Fishermen and Fishing Vessels, 2005





Air pollution

- MARPOL Annex VI
  - adopted 1997, entered into force: May 2005
  - Revised Annex VI adopted 2008, entry into force 2010
  - Progressive reductions in SOx and Nox,
  - Stricter controls in ECAs



Greenhouse gas emissions
Energy efficiency measures for ships adopted under MARPOL Annex VI

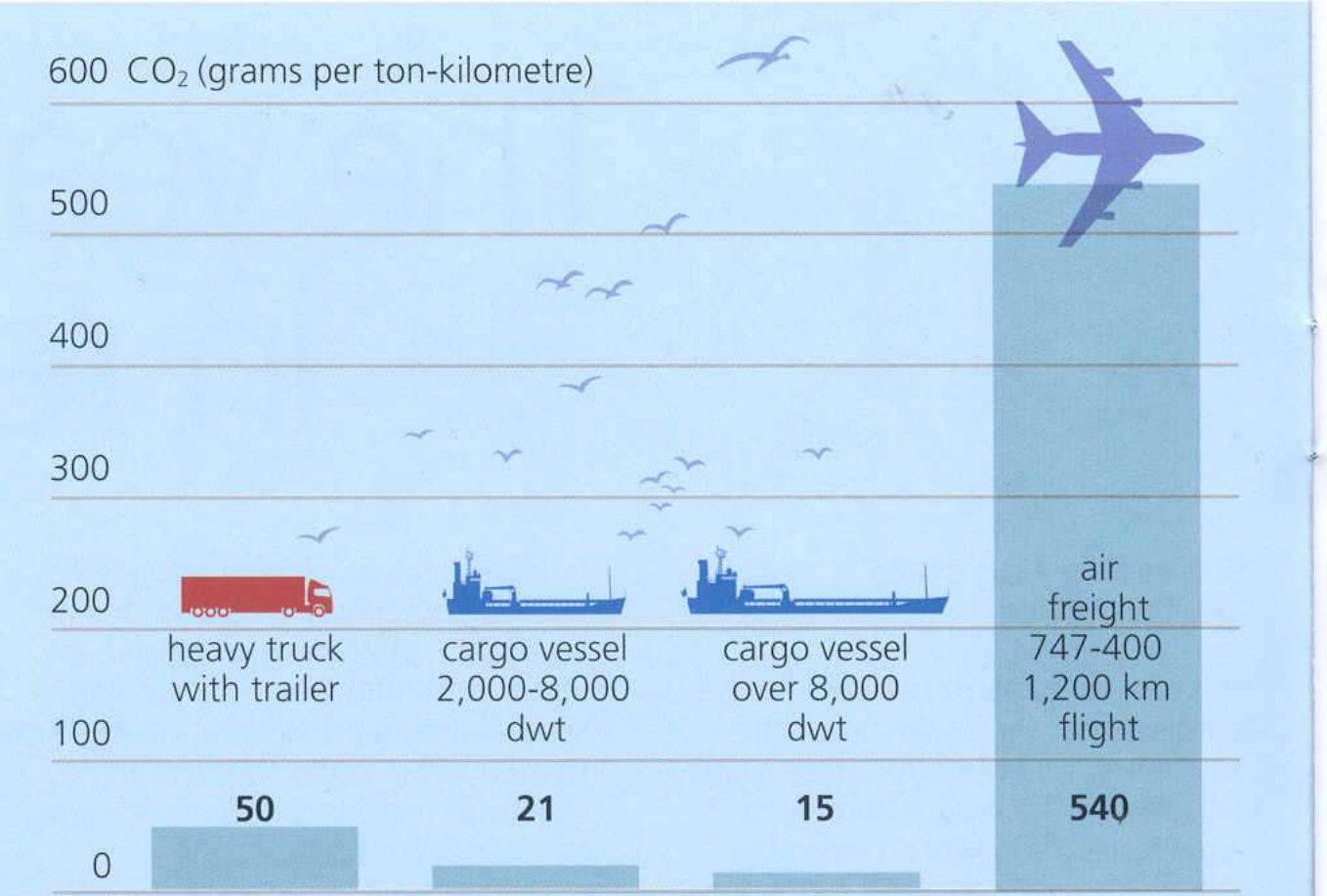


- Technical and operational measures adopted 2011, entered into force 2013

- Energy Efficiency Design Index (EEDI) for new ships
  - intended to stimulate innovation and technical development of all the elements influencing the energy efficiency of a ship from its design phase

- Ship Energy Efficiency Management Plan (SEEMP)
  - mandatory for new and existing ships
  - SEEMP should incorporate best practices for the fuel efficient operation of ships

Carbon footprint?



Comparison of CO_2 emissions by different transport modes

Source: NTM (Swedish Network for Transport and the Environment)

Alien invaders in ballast water

- Problem – brought to IMO in 1988
  - micro-organisms transported to alien eco-systems in ships' ballast water, no predators; environmental damage; enter food chain







- Solution
  - International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004

    GloBallast prgramme - collaboration between IMO, GEF, UNDP, governments
    and industry to assist less-industrialised countries tackle ballast water problem

Ship recycling

- Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009

- Environmentally friendly disposal of old ships:
  - Most components and materials re-used
- But: safety issues for workers - hazardous materials
- Inter-agency co-operation ILO, Basel Convention







The human element

Complex, multi-dimensional issue

2010 – conference adopted revised STCW Convention

International Day of the Seafarer – 25 June social media campaigns





Memorial to Seafarers

- Major work of sculpture
- Unveiled 2001
- Reminder to staff, delegates and visitors of seafarers' role



Does it work? Lives lost….

Definition: (a) Number of lives lost (seafarers, fishers and passengers) due to safety-related[1] accidents and incidents on ships subjec Conventions and other instruments.

i.e. not accidents and incidents which are due to security failures, acts of piracy and armed robbery or whose prevention is addressed by other international Conventions.

Source: IHS Fairplay (merchant vessels over 100 GT).
Note: The Secretariat has records of data on incidents involving fishing vessels but insufficient data provided via GISIS on fishermen's' lives lost

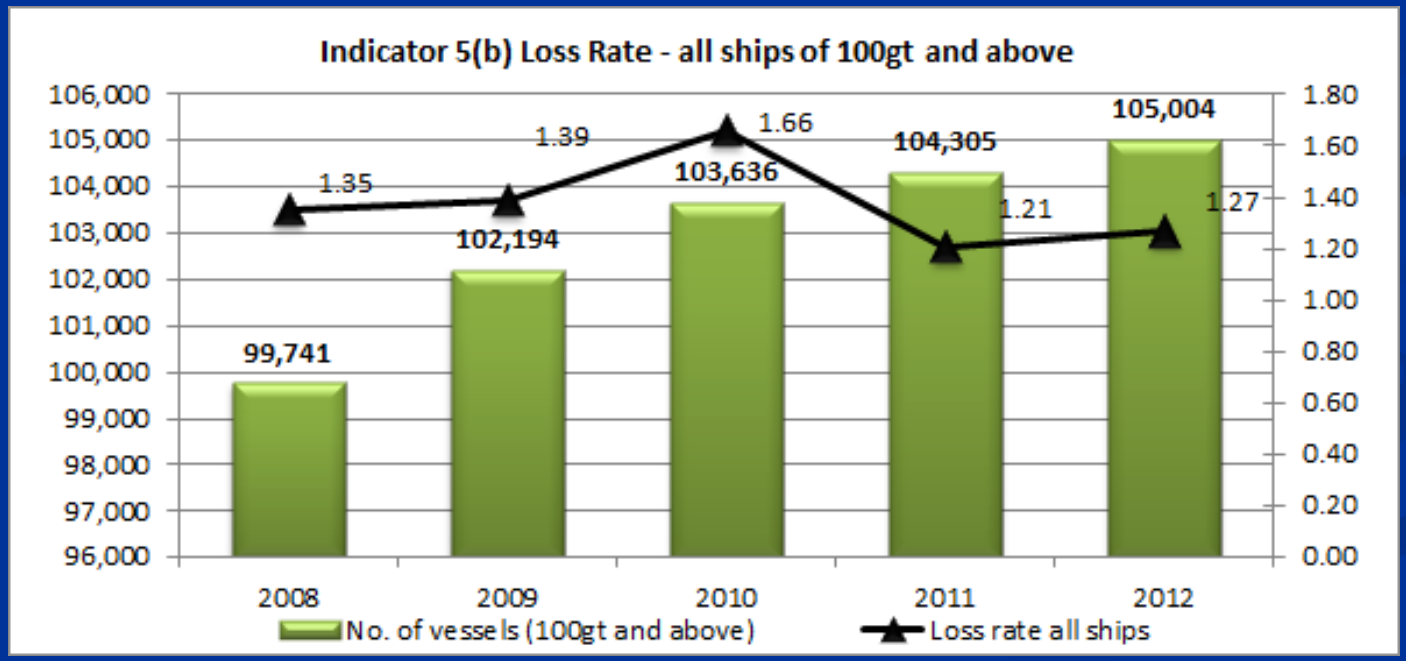
Total amount of passengers and crew 2012: 2,201,120,049
Ratio estimate per 1000 lives: 0.00028

	2008	2009	2010	2011	2012
IHS Fairplay data	1,160	699	250	3,214	610
IMO	1,921	2,395	1,622	1,145	1,401

Ship losses over the years - declining

- 1966 to 1985: more than 300 ships lost annually.
- 1990: under 200; 2000: 167 lost.

- 2,596 shipping casualties reported in 2013, only 94 were considered total losses, according to Allianz.



Shipping affects us all – sustainable development

- Nearly 90% of world trade carried by sea
  - Raw materials and commodities
  - Finished goods
  - Foodstuffs
  - Fuel
- Underpins global economy
- Safe, secure and environmentally friendly transport system







RECOMMENDED that the Maritime Safety Committee:

.1 consider how to improve data collection and increase its availability in order to support monitoring and development of safety regulations;

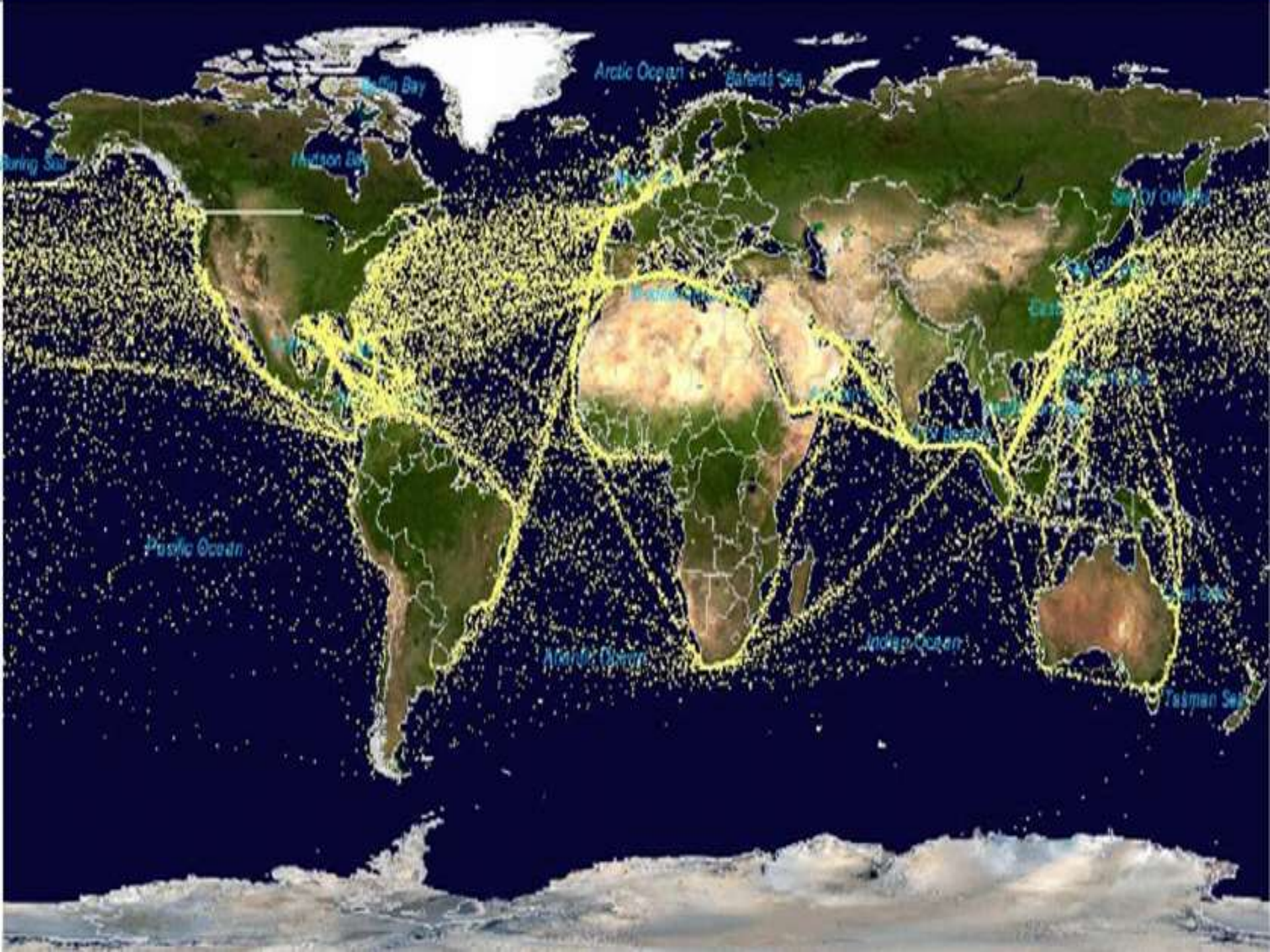
.2 consider how to better integrate risk-based methodologies and the latest analysis techniques into the safety regulatory framework to provide a sound scientific and practicable basis for the development of future safety regulations;

.3 consider ways of encouraging a safety culture beyond mere compliance with regulatory requirements;



.4 **take into account the burden any new or changing regulation(s) place on the seafarers and consider how this burden can be minimized;**

.5 consider undertaking a long-term comprehensive review of the existing safety regulatory framework with a view to ensuring that it will meet the future challenges associated with the application of new technologies, the human element, the needs of the maritime industry and the expectations of society, taking into account the ever-increasing pace of change and technological advancements made since the 1974 SOLAS and the International Load Lines Conventions were adopted.


Arctic Ocean
Barents Sea
Pacific Ocean
Atlantic Ocean
Indian Ocean
Tasman Sea

International Maritime Organization

Safe, secure and efficient shipping on clean oceans



Essential Elements of Article 8*bis*

- Requirement to cooperate to the fullest extent possible
- Process & options for making, receiving, and responding to boarding requests
- Allocation & preservation of enforcement jurisdiction
- Conduct of boarding & disposition operations
- Recourse for damage, harm, or loss
- Further implementation